

On-Prem to cloud Migration using Azure Migrate



BY: Shivam G. Joshi

Pre-requisites:

- An existing Hyper-v server to migrate
- Azure subscription(Active)

Migration Phase



Steps of Migration:

Decide: The first step in a migration journey is discovering your workloads. You can do this by deploying a lightweight Azure Migrate appliance (recommended) or by quickly importing inventory data. This process of identification is called *discovery*.

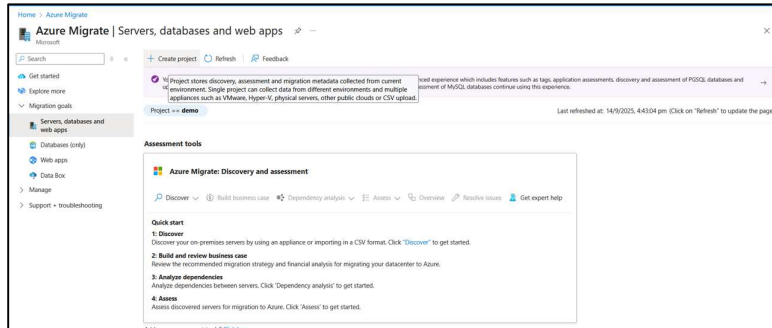
Plan: Once you've decided to migrate to Azure, the next key phase is detailed planning. Use Azure Migrate assessments to help with:

- **Azure readiness:** Check if your on-premises servers, SQL servers, and web apps are ready for Azure.
- **Right-sizing:** Estimate the appropriate Azure VM sizes, SQL configurations, and the number of Azure VMware Solution nodes needed.
- **Cost estimation:** Calculate the projected cost of running your current workloads in Azure.
- **Dependency analysis:** Identify network dependencies between servers to ensure a smooth migration without overlooking critical connections.

Execute: In this phase, you carry out the migration or modernization of your workloads to Azure. Use Azure Migrate or partner tools to move servers, databases, web apps, or virtual desktops with minimal downtime and reduced risk.

Lets get into practical mode:

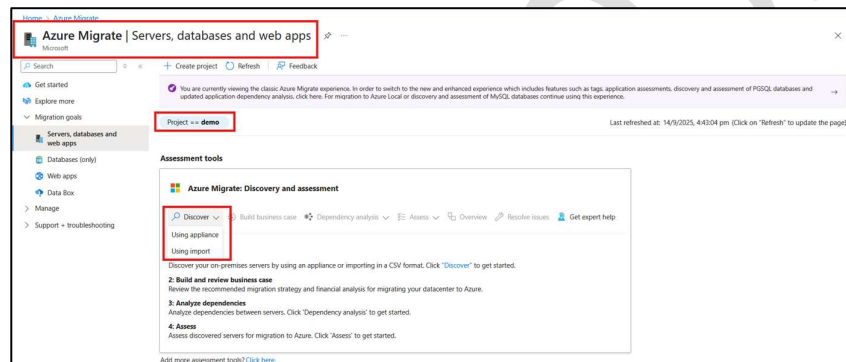
1- Login to the Azure Portal -> In the search bar, type **Azure Migrate** and select it ->Click **Create project** -> Select **Subscription**, **Resource Group**, enter **Project name**, and choose the **Geography** -> Click **Create**.



2- Go to your **Azure Migrate Project** -> Click on **Discover**.

You will get two options for discovery:

- **Using Appliance** → Deploy an appliance in your environment to automatically discover machines.
- **Using Import** → Import a CSV file with server details for assessment.



When to use which option:

Using Azure Migrate Appliance

- **What it is:**
A lightweight virtual appliance (VM) you deploy on-premises (VMware/Hyper-V/physical).
- **How it works:**
 - Collects **real-time metadata** and **performance data** (CPU, memory, disk, network) from your servers.
 - Sends that data securely to Azure Migrate.
- **Benefits:**
 - **Automatic discovery** of machines (no manual entry).
 - **Dependency mapping** (sees which servers talk to each other).
 - **Performance-based sizing** (Azure VM recommendations based on actual utilization, not just configuration).
 - **Ongoing updates** (continuous data collection).
- **Use case:**
Best for **large environments** or when you need **accurate right-sizing** and **dependency visualization**.

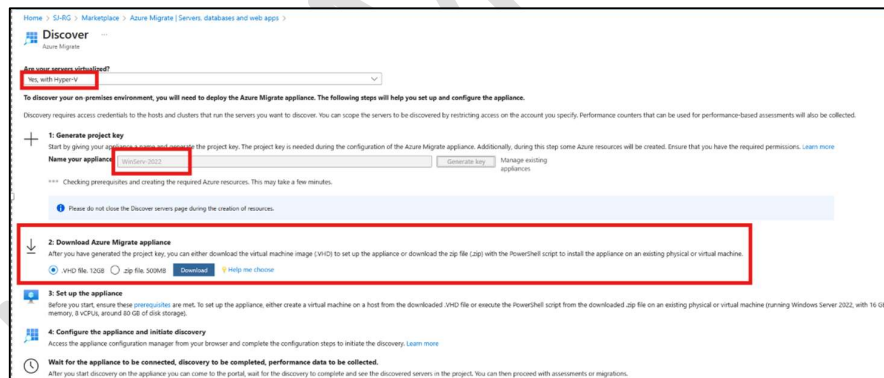
Using Import (CSV file)

- **What it is:**
A manual method where you prepare a CSV with details of your servers.
- **How it works:**
 - You export inventory data from tools (like CMDB, vCenter, SCCM) or manually collect it.
 - Upload the CSV into Azure Migrate.
- **Benefits:**
 - **Quick and simple** (no need to deploy an appliance).
 - Good for **small environments** or **initial rough assessment**.
- **Limitations:**
 - No performance data → only **as-is sizing** (based on allocated CPU/RAM, not actual usage).
 - No automatic discovery or dependency mapping.
 - Static snapshot — you need to re-import if things change.
- **Use case:**
Best for **small migrations** or when security rules prevent deploying an appliance.

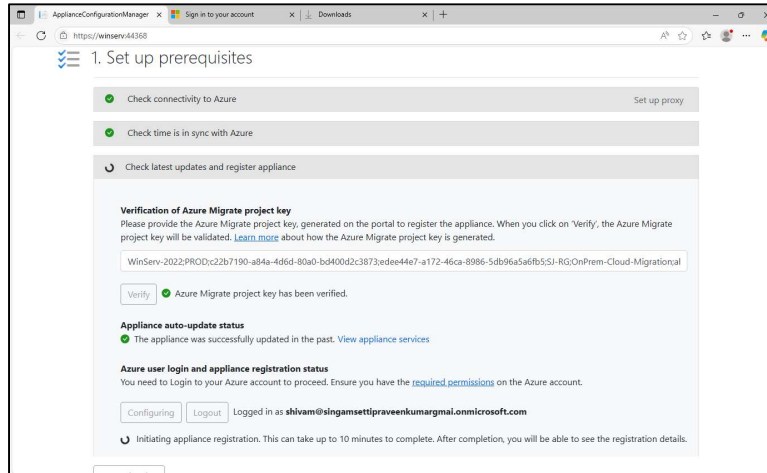
In short:

- **Appliance** = best for accuracy, performance-based recommendations, and larger projects.
- **Import** = best for quick, simple assessments when you don't want to deploy anything.

3- Select the **Server Virtualization Type** (e.g., VMware, Hyper-V, Physical) -> Enter a **name for your appliance** -> Click **Generate Key** → copy and save this key securely (it will be required in later steps during appliance setup).



4- Once the **key** is generated, copy the **download link** provided -> Download the **.vhd file** from the link -> Attach the downloaded **.vhd file** to your **Hyper-V server** as a disk -> Start the appliance VM → it will automatically open a browser window on your Hyper-V server at **port 44368**.



5- At this stage, you may face an issue where the **appliance VM does not have internet connectivity** -> To resolve this, create a **Virtual Switch / Network Adapter** in Hyper-V -> Attach the appliance VM to this new virtual switch so it can share the internet connection from your Hyper-V server -> Verify that the appliance VM can access the internet before proceeding further.

6-Follow these steps to establish internet connection to your Hyper-V machine.

1-Give an IPv4 address and a gateway to your Hyper-V server

2-[Run Hyper-V in a Virtual Machine with Nested Virtualization | Microsoft Learn](#)

[nested hypervisors on Azure VM - Microsoft Q&A](#)

3- Open **Hyper-V Manager** on your server -> Click on **Virtual Switch Manager** -> Select **New Virtual Network Switch** -> choose **External** -> Bind the switch to the **physical network adapter** that has internet access -> Assign this **External Switch** to your Appliance VM's network adapter -> Start the Appliance VM and confirm that it now has **internet connectivity** (try browsing or pinging an external site).

- You can use Azure DNS IP or Google DNS IP. Once this is done try to browse a public site for testing.

7- Now, refer to the screenshot above and paste the **key** that was copied earlier from the Azure portal. Allow the system to verify it; once verified, the appliance will also update automatically.

Next, proceed to log in with your **Azure account**:

- Click **Login**, copy the **device code**, and enter the Azure-registered **email ID**.
- Paste the code, provide your credentials, and complete the sign-in.

Once the account is successfully verified, the appliance setup is complete and ready to use.

8-Now we need to provide our host credentials in the below section:

2. Manage credentials and discovery sources

Step 1: Provide Hyper-V host credentials for discovery of Hyper-V VMs

You can provide multiple credentials for Hyper-V hosts. [Learn more](#) about the permissions required on Hyper-V host credentials.

[Add credentials](#)

Search:

#	Source type	Friendly name	Username	Actions
1	Hyper-V	OnpremCred	.\Administrator	Edit
2	Hyper-V	OnpremCredadmin	.\adminuser	Edit
3	Hyper-V	azureuserhost	.\azureuser	Edit

Provide a **friendly name** to easily identify the credentials, then enter the **username** and **password** of the host server.

9- Once the credentials are saved, the next step is to provide the **source details** — specifically, the **Hyper-V server information** that needs to be migrated.

Step 2: Provide Hyper-V host/cluster details

You can add both Hyper-V host and clusters as discovery sources. If you add a host that is a part of a cluster, the cluster is detected and added to the list. For a successfully validated Hyper-V host/cluster, you can view more details by clicking on its IP Address/ FQDN in the table below. [Learn more](#) about the prerequisites for Hyper-V VMs discovery.

☐ Enabling this slider will enforce the HTTPS protocol when the appliance connects to target servers. When the [pre-requisites](#) are not met on the target servers, discovery will fail on both the currently discovered and newly added target servers.

When the slider is disabled, the default communication is over the HTTPS protocol. When the [pre-requisites](#) are not met on any of the target servers, the communication falls back to the HTTP protocol.

[Add discovery source](#)

Search:

Filter status: All

#	Source type	Mapped credentials	IP Address / FQDN	Port	Status	Delete
1	Hyper-V Host	azureuserhost	SJ-VM	5986/5985	Validation successful	Delete

Click on **Add discovery points** and provide the **IP address or hostname** of the server.

- The initial validation may fail because the required firewall ports (**5986/5985**) are not open on the host machine.
- If validation fails, copy the provided link and run it on the **Host server** using **PowerShell**.
- This will download and execute a script that installs the missing dependencies and adds the necessary firewall rules automatically.

10-Now we are starting the actual discovery of our inventory.

Search: Search based on Credentials type, Friendly name, Username or Status

Filter status: All

#	Credentials type	Friendly name	Username	Status	Actions
1	Windows (Non-domain)	OnpreCred	Administrator	Not applicable	Edit Delete

[Edit SQL Server connection properties.](#)

You can revalidate the added domain credentials by clicking on the button below.

[Revalidate credentials](#)

Click on the button below to initiate discovery. After the discovery is complete, you can check the discovery status of the Hyper-V hosts in the table above. [Learn more](#) about the metadata collected during discovery.

[Start discovery](#)

Discovery has been successfully initiated. Go to the [Azure portal](#) to review the discovered inventory.

Once the details are filled hit start discovery.

11- Once the discovery process is complete, navigate to the **Azure Portal** → go to **Discovered Items**. Here, you will be able to see the details of your **on-premises Hyper-V** servers that have been successfully discovered.

Home > Azure Migrate

Azure Migrate | Discovered items

Search: Create group Assess Export Data Dependency analysis Enable Arc Refresh Columns Feedback

You are currently viewing the classic Azure Migrate experience. In order to switch to the new and enhanced experience which includes features such as tags, application assessments, discovery and assessment of PostgreSQL databases and updated application dependency analysis, click here. For migration to Azure Local or discovery and assessment of MySQL databases continue using this experience.

Subscription: Azure subscription Project: Onprem-Cloud (S...)

Microsoft tools

To view the machines discovered by the Azure Migrate appliance, click on the 'Azure Migrate appliance' tab. For all others machines, click 'Other'.

Azure Migrate appliance Import based Other

Appliance name: onpremapplian (Hyper-V) Workload: All servers

Search to filter rows

Name	IPv4/IPv6	Source	Support status	Software inventory	Dependencies (Age...	Database instances	Web app	Arc status	File Serv
OnpreVM	192.168.100.21	SI-VM	-	19 Applications	Not enabled	0	2	Unknown	No

Previous Page 1 of 1 Next

Here our discovery steps gets completed.

12- Now proceed to the **Assessment** phase.

This step helps analyze the discovered servers and provides recommendations to finalize the appropriate **compute size** and **Azure region/location** that will be used for the target server after migration.

Home > Azure Migrate

Azure Migrate | Discovered items

Search: Create group Assess Export Data Dependency analysis Enable Arc Refresh Columns Feedback

You are currently updated appo...

Subscription: Azure MySQL database Project: Onprem-Cloud (S...)

Microsoft tools

To view the machine Azure VMware Solution (AVS) appliance, click on the 'Azure Migrate appliance' tab. For all others machines, click 'Other'.

Azure Migrate appliance Import based Other

Appliance name: onpremapplian (Hyper-V) Workload: All servers

Search to filter rows

Name	IPv4/IPv6	Source	Support status	Software inventory	Dependencies (Age...	Database instances	Web app	Arc status	File Serv
OnpreVM	192.168.100.21	SI-VM	-	19 Applications	Not enabled	0	2	Unknown	No

Previous Page 1 of 1 Next

Select the above fields as per the requirements.

13-Create assessment and select the discovered server.

Home > Azure Migrate | Discovered items >

Create assessment

Assessment name * OnpremServer-Assessment

Select or create a group

Create new Use existing

Group name * AssessmentGroup

Add machines to the group

Appliance name * onpremsaplan (Hyper-V)

Select all Clear selection Search to filter machines

Name	Issues	IPv6/IPv4	Operating system	Operating system support st...	Support ends in	Source	Machine type
☐ OnpremVM	Details	192.168.100.21;680-3dc4a04...	Windows Server 2022 Enterprise	Mainstream	74 months	SI-VM	Hyper-V

Once the assessment is completed, you will receive the assessment report. In some cases, it may take a while to generate the detailed report.

14-Below are some examples of Assessment report.

Home > Azure Migrate | Discovered items >

Create assessment

Assessment name * OnpremServer-Assessment

Select or create a group

Create new Use existing

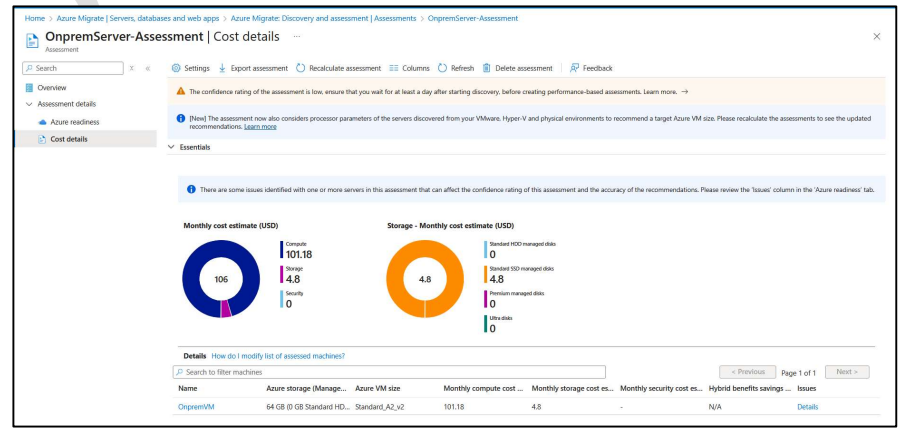
Group name * AssessmentGroup

Add machines to the group

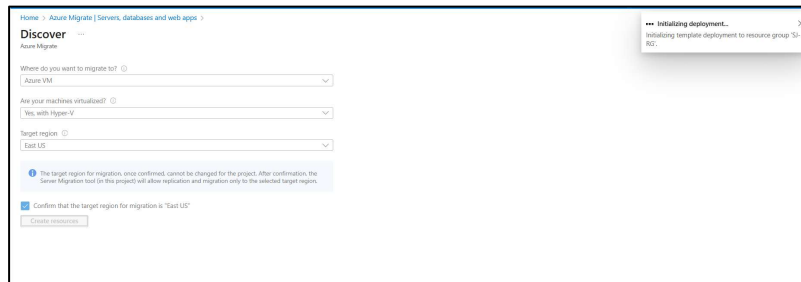
Appliance name * onpremsaplan (Hyper-V)

Select all Clear selection Search to filter machines

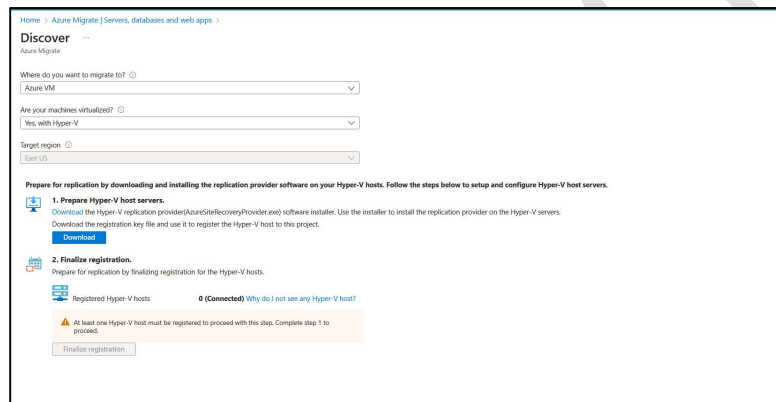
Name	Issues	IPv6/IPv4	Operating system	Operating system support st...	Support ends in	Source	Machine type
☐ OnpremVM	Details	192.168.100.21;680-3dc4a04...	Windows Server 2022 Enterprise	Mainstream	74 months	SI-VM	Hyper-V



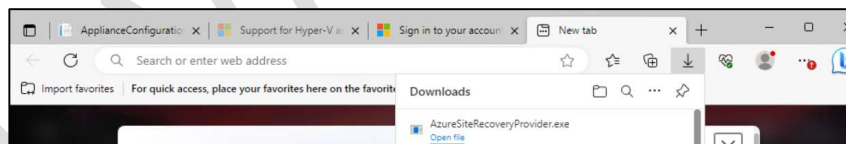
15- After deploying the appliance in your Hyper-V environment, you need to register the site with Azure Migrate. This process creates a Hyper-V site, serving as a logical container for your discovered Hyper-V hosts and virtual machines.



16- After the resources are created in the target region, a download button will appear to download AzureSiteRecoveryProvider.exe. This file is used to install the replication provider on Hyper-V servers. You will also receive a key file that must be used when running the .exe file.



17-Download the .exe file



18- Open the .exe file and browse to select the key file. After providing the key, it will automatically retrieve information about your Azure subscription and related details.

Microsoft Azure Site Recovery Registration Wizard

Vault Settings...

Select the registration key file you downloaded from the Azure Site Recovery portal and specify vault settings. [Learn More](#)

Key file	Onprem-Cloud-MigrateVault-174496451_Onprem-Cloud-HyperVSite_V	Browse
Subscription	ec7d4d49-5537-4d5e-affc-d99f9125c968	
Vault name	Onprem-Cloud-MigrateVault-174496451	
Hyper-V site name	Onprem-Cloud-HyperVSite	

Next Cancel

19-Once this is done your server gets registered.

Microsoft Azure Site Recovery Registration Wizard

Registration

✓ The server was registered in the Azure Site Recovery vault

Finish

20-If you check on the portal level, you can see count of registered servers.

Home > Discover

Where do you want to migrate to? Azure VM

Are your machines virtualized? Yes, with Hyper-V

Target region: East US

Prepare for replication by downloading and installing the replication provider software on your Hyper-V hosts. Follow the steps below to setup and configure Hyper-V host servers.

- 1. Prepare Hyper-V host servers.**
Download the Hyper-V replication provider (AzureSiteRecoveryProvider.exe) software installer. Use the installer to install the replication provider on the Hyper-V servers.
Download the registration key file and use it to register the Hyper-V host to this project.
[Download](#)
- 2. Finalize registration.**
Prepare for replication by finalizing registration for the Hyper-V hosts.
Registered Hyper-V hosts: 1 (Connected)
Why do I not see any Hyper-V host?
Registration finalized

Notifications

More events in the activity log →

Deployment succeeded

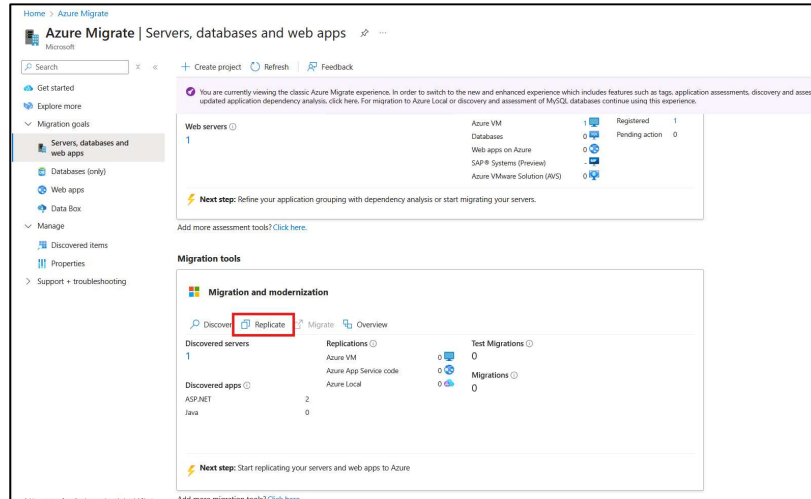
Deployment: Microsoft.MigrateV2.HyperVSite.PolicyCreationAndAssociation-202 to resource group 'SI-RC' was successful.

[Pin to dashboard](#) [Go to resource group](#)

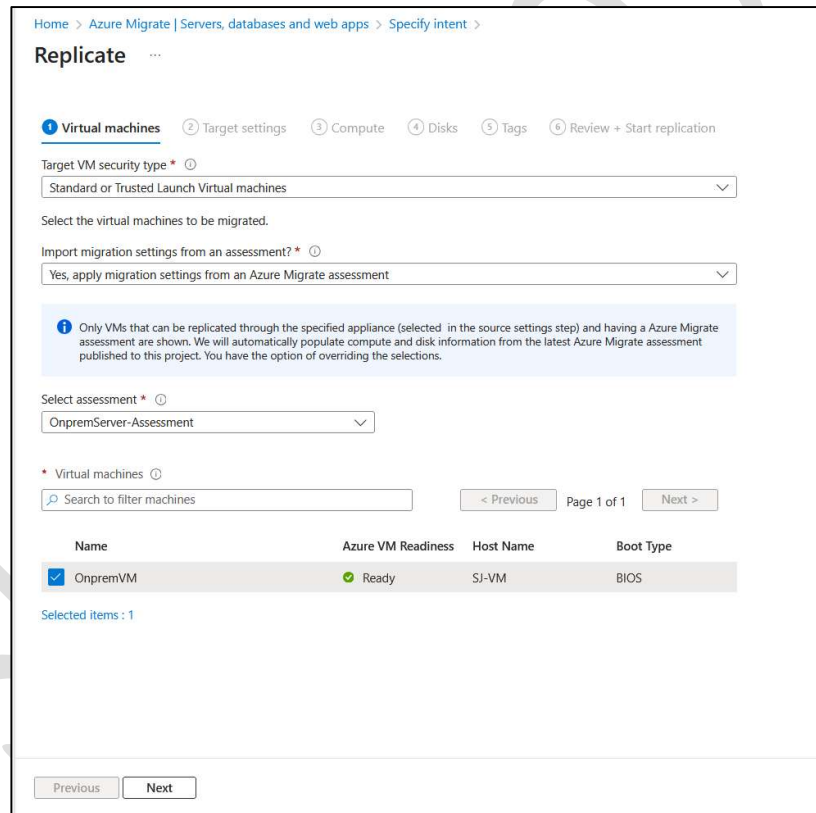
a few seconds ago

Close Feedback

21-Now replicate the server.



22- Here, specify the security type required for your server in Azure, import the migration settings from the assessment, and include the discovered server.



23- Once you initiate the server replication, please allow some time for the replication status details to be updated and reflected.

Home

Site Recovery jobs

Onprem-Cloud-HyperVSite

Filter Export jobs

Filter items...

Name	Status	Type	Item	Start time	Duration
Enable replication	Successful	Protected item	OnpremVM	9/10/2025, 6:03:57 PM	00:01:17
Associate replication policy	Successful	Replication policy	defaultOnprem-Cloud-Hyper...	9/10/2025, 5:52:50 PM	00:02:37
Create replication policy	Successful	Replication policy	defaultOnprem-Cloud-Hyper...	9/10/2025, 5:52:13 PM	00:00:00
Register the Azure Site Recov...	Successful	Server	SI-VM	9/10/2025, 4:54:18 PM	00:00:00
Create a site	Successful	Server	Onprem-Cloud-HyperVSite	9/10/2025, 2:50:04 PM	00:00:00

Notifications

More events in the activity log →

Dismiss all

Starting server replication

Successfully completed the operation.

4 minutes ago

Deployment succeeded

Deployment: Microsoft.MigrateV2.ArmTemplateCreation-202581018237 to resource group 'SI-RG' was successful.

Pin to dashboard Go to resource group

6 minutes ago

Deployment succeeded

Deployment: Microsoft.MigrateV2.HyperVSite_PolicyCreationAndAssociation-202 to resource group 'SI-RG' was successful.

Pin to dashboard Go to resource group

14 minutes ago

24-Once you get the below out put you are ready for migration.

Home > Azure Migrate | Servers, databases and web apps > Azure Migrate: Server Migration

Azure Migrate: Server Migration | Replications

Search Refresh Migrate Columns Feedback

Overview Migration

Replications

Infrastructure servers

Jobs

Events

Azure Local migration

Replications

Jobs

Events

Settings

Properties

Support + troubleshooting

Replication using classic experience for agent-based VMware and physical server migrations will be deprecated by 30 March 2026. Plan your migrations accordingly. Use simplified experience. [Learn more about simplified experience](#)

Other

Last refreshed at: 11/9/2025, 9:26:58 am

Search to filter items

Name	Replication status	Replication health	Test migration status	Migration status
OnpremVM	Waiting for first recovery point	Healthy	Never performed	-

Showing 1 - 1 of 1 results.

Home > Azure Migrate | Servers, databases and web apps > Azure Migrate: Server Migration

Azure Migrate: Server Migration | Replications

Search Refresh Migrate Columns Feedback

Overview Migration

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Azure Local migration

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Support + troubleshooting

Replication using classic experience for agent-based VMware and physical server migrations will be deprecated by 30 March 2026. Plan your migrations accordingly. Use simplified experience. [Learn more about simplified experience](#)

Other

Last refreshed at: 11/9/2025, 9:38:12 am

Search to filter items

Name	Replication status	Replication health	Test migration status	Migration status
OnpremVM	Protected	Healthy	Never performed	-

Showing 1 - 1 of 1 results.

Test migration

Clean up test migration

Migrate

Resynchronize

Health error details

Stop replication

25- Navigate to the Azure Migrate homepage, then select Server, DB, and Web Apps. From there, go to Server Migration and click on Migrate.

Home > Azure Migrate | Servers, databases and web apps > Azure Migrate: Server Migration | Replications >

Migrate

OnpremVM

Name

OnpremVM

Operating system

Version unknown

Check for upgrade

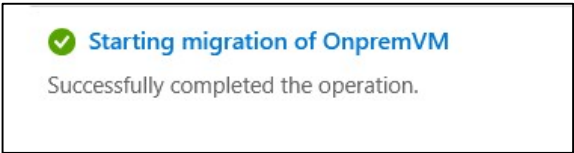
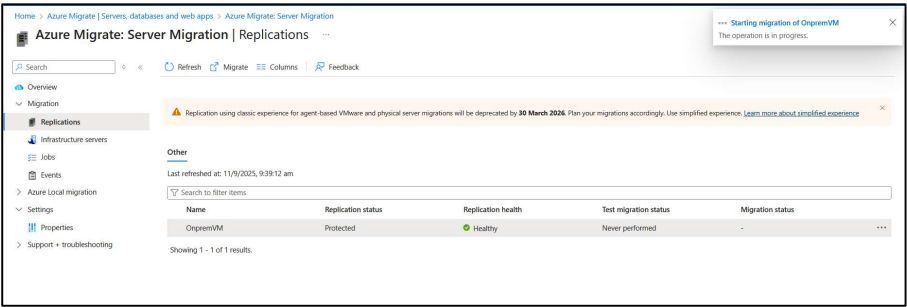
Shutdown virtual machines and perform a planned migration with no data loss? *

Yes

Migrate

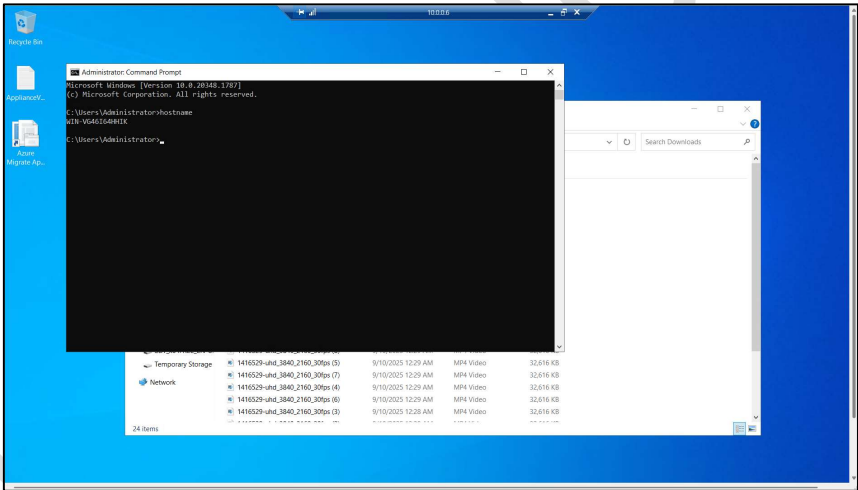
Feedback

Migration in progress



26- After the migration is completed, you will notice that your Hyper-V server is now in a stopped state.

27- Final Output. Now my onprem hyper-v server is now migrated to Azure cloud.



Useful Links:

Azure Migrate: <https://learn.microsoft.com/en-us/azure/migrate/migrate-services-overview?view=migrate-classic>

Azure Migrate Appliance: <https://learn.microsoft.com/en-us/azure/migrate/migrate-appliance?view=migrate-classic>

Appliance Security Best Practice: <https://learn.microsoft.com/en-us/azure/migrate/best-practices-security?view=migrate-classic>

Confidence rating: <https://learn.microsoft.com/en-us/azure/migrate/confidence-ratings?view=migrate>

Hyper-v Migration to Cloud: <https://learn.microsoft.com/en-us/azure/migrate/tutorial-migrate-hyper-v?view=migrate-classic&tabs=UI>

Creating Business Case: <https://learn.microsoft.com/en-us/azure/migrate/concepts-business-case-calculation?view=migrate-classic>